Exercise 5.16

An NMOS transistor has $V_{t0} = 0.8$ V, $\phi_f = 0.35$ V, and $\gamma = 0.4\ V^{1/2}$.
Find V_t when $V_{SB} = 3V$.

$$\Delta V_t = \gamma\left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F}\right)$$

$$= 0.4V^{\frac{1}{2}}\left(\sqrt{0.7V + 3} - \sqrt{0.7}\right) = 0.43V$$

$$V_t = 0.8V + 0.43V = 1.23V$$